SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Operating Systems: Process Synchronization & Memory Management

Department: CSE	**Course / Subject:** Operating Systems (CS403PC)
Semester: Semester 4	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Process control blocks, CPU scheduling, Semaphores & Monitors, Banker's Deadlock Algorithm, Demand Paging, Virtual Memory, and File System structure.

Unit 1: Process Management & Scheduling

Process states, Process Control Block (PCB), Context Switching, FCFS, SJF, Priority Scheduling, Round Robin, and Multi-level Queue Scheduling.

Unit 2: Inter-Process Communication & Synchronization

Race conditions, Critical section problem, Peterson's algorithm, Counting & Binary Semaphores, Classic IPC problems (Dining Philosophers, Readers-Writers).

Unit 3: Deadlocks

Deadlock characterization (4 Coffman conditions), Resource Allocation Graphs, Deadlock prevention, Banker's safety algorithm, and Deadlock detection & recovery.

Unit 4: Memory Management & Paging

Logical vs physical addresses, Paging hardware, Translation Lookaside Buffer (TLB), Segmentation, Demand paging, Page replacement algorithms (FIFO, LRU, Optimal).

Unit 5: File & Disk Management

File allocation methods (Contiguous, Linked, Indexed), Free space management, Disk scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK).